

Srinivasan Poonkundran

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PROFESSIONAL EXPERIENCE

Cognizant Technology Solutions

Sept. 2021 – Jan. 2024

Full Stack Engineer

Remote

- Developed responsive web interfaces across 3 enterprise applications, supporting close to 15,000 active monthly users on both desktop and mobile platforms.
- Optimized application performance, reducing average page load time from 4 seconds to 3 seconds by implementing caching strategies and rewriting SQL queries, improving experience for nearly 10,000 daily users.
- Redesigned 20+ UI components in Angular and TypeScript, which increased the average session duration from 3 minutes to 3.5 minutes and reduced bounce rates by approximately 1,200 users per month.
- Built a Python-based PDF label validator that eliminated about 150 hours of manual checks annually, saving \$1,600 in labor costs and reducing average processing time per document from 2 minutes to under 10 seconds.
- Created and optimized over 100 SQL queries and 25 stored procedures for reporting and application logic, cutting average data retrieval time from 8 seconds to under 5 seconds.
- Built interactive dashboards in Tableau and Power BI to visualize KPIs such as order volume, SLA breaches, and customer churn, enabling business teams to track performance across 6 regions.
- Automated generation of Excel/CSV-based performance reports using Python and Pandas, reducing manual reporting time from 4 hours per week to under 30 minutes.
- Collaborated in Agile teams of 10–12 members across 12+ quarterly release cycles, contributing to sprint planning, backlog grooming, and code reviews that closed nearly 250 Jira tickets per quarter.
- Mentored 3 junior developers through weekly code reviews and pair programming sessions, reducing the number of code review defects per sprint from 25 to 20.
- **Skills:** Java, Spring Boot, Unit Testing, Angular, Python, AWS, MySQL, Microservices, CI/CD, Performance Optimization, Problem Solving, Data Analytics & Visualization, Docker, Agile, Cross-functional Collaboration

Clothing Store Billing and Management System – Local Retail Client

Jan. 2023 – Jan. 2024

Software Developer

Remote

- Developed a full-featured Android application to manage invoices, sales, inventory, and staff operations for a retail clothing store handling more than 1,000 monthly transactions.
- Designed and deployed interactive sales dashboards with visual charts to track daily revenue, top-selling items, seasonal demand trends, and customer purchase history, improving business decision-making for 3 store managers.
- Implemented barcode-based billing to automate checkout processes which reduced average billing time from 3 minutes to 90 seconds per transaction and cut manual entry errors from 20 per week to fewer than 1.
- Integrated WhatsApp API for automated invoice sharing, enabling instant digital receipts to 700+ recurring customers and reducing paper invoice costs by \$500 annually.
- Configured AWS S3 cloud storage to securely back up over 12,000 invoices and sales records, providing cross-device access and reducing risk of local data loss by 100%.
- Built an SQLite database to handle customer profiles, inventory items (nearly 5,000 SKUs), and staff details, supporting real-time updates and reporting across multiple devices.
- Developed automated reporting modules to generate weekly and monthly summaries of revenue, profit margins, and stock movements, eliminating 8+ hours of manual report preparation per month.
- Collaborated with the client remotely through weekly demo sessions, collected feedback, and delivered 4 incremental releases over 12 months using Agile methodology.
- Enhanced application security by encrypting customer and transaction data during storage and transfer, meeting compliance requirements for digital invoicing.
- **Skills:** Android, Java, AWS S3, SQLite, Data Visualization, WhatsApp API, UI/UX Design, RegEx, Agile Development

INTERNSHIP EXPERIENCE

Xerxez

Oct. 2020 — Dec. 2020

Cloud Computing & Machine Learning Intern

Remote

- Completed a 6-week internship focused on Cloud Computing and Machine Learning using Amazon Web Services (AWS), gaining hands-on experience with cloud infrastructure and project deployment.
- Assisted in developing machine learning models and deploying them on AWS, enhancing understanding of cloud-based data workflows.

- Collaborated with the Project Division team on AWS-based ML projects, learning best practices for cloud project management and data processing pipelines.
- Demonstrated initiative and curiosity by exploring advanced ML concepts and AWS services beyond the assigned tasks.
- **Skills:** AWS, Cloud Computing, Machine Learning, Python, Data Analysis, Project Collaboration

LEADERSHIP EXPERIENCE

Acadflip – EdTech Startup (“The Book of Video Lectures”)

Jan. 2020 – Sept. 2020

Brand Ambassador

Remote

- Appointed as an official Brand Ambassador of Acadflip, representing the platform among students and educators.
- Promoted Acadflip’s video lecture platform on campus and social media, reaching an audience of nearly 500 peers.
- Organized awareness campaigns and peer learning sessions to drive student engagement and increase platform adoption.
- Acted as a liaison between students and the startup team, providing feedback that helped improve product features and user experience.
- **Skills:** Campus Outreach, Public Speaking, Social Media Promotion, Community Engagement

PROJECTS

GradeLens - RAG Powered Auto Grader | [GitHub](#)

- Developed a web-based grading platform leveraging Retrieval Augmented Generation (RAG) to automate student paper evaluation using professor-uploaded lecture content in PDF and PPT formats.
- Designed the RAG core and grading logic to ensure grades are context-aware, consistent, and aligned with course material.
- Implemented a dynamic web module for rubric management, allowing professors to add, edit, or remove rubric criteria for flexible evaluation.
- Built backend services with Django and Django REST framework for file ingestion, rubric application, and grading results, ensuring scalable data flow.
- Developed React based front-end dashboard for professors to upload course content, manage classes, and review student grades efficiently.
- Achieved 90% grading accuracy compared to human evaluators, with automated feedback generated faster than manual grading.
- **Skills:** Retrieval Augmented Generation (RAG), Hugging Face, Natural Language Processing (NLP), Large Language Models (LLMs), Python, Django + Django REST framework, React, Document Ingestion

Emotion Detection from Text | [GitHub](#)

- Developed a Natural Language Processing (NLP) system to classify English text into 7 emotions (anger, joy, sadness, fear, surprise, disgust, neutral), extending sentiment analysis beyond traditional polarity detection.
- Implemented a BERT-based embedding model (bert-base-uncased), freezing transformer layers during training to reduce GPU resource usage while maintaining accuracy.
- Built a deep learning classifier in Keras using Bidirectional GRUs, dropout regularization, and dense layers, achieving an F1-micro score of 0.83 on the validation dataset.
- Optimized training with TensorFlow techniques including mixed precision, ModelCheckpoint, and EarlyStopping, reducing training time by nearly 30%.
- Preprocessed large-scale text data using Hugging Face libraries and applied hyperparameter tuning (learning rate, batch size, GRU units) to improve model generalization.
- Created an end-to-end inference pipeline for real-time prediction, with result formatting to support downstream applications such as chatbots and feedback analysis.
- **Skills:** BERT, Hugging Face, TensorFlow, Keras, GRU, Natural Language Processing (NLP), Python

Heart Rate Monitoring System | [GitHub](#)

- Developed an IoT-enabled web application to monitor heart rate and blood oxygen (SpO2) levels in real time, supporting continuous health tracking for end-users.
- Implemented a backend with Node.js, Express, and MongoDB to design secure RESTful APIs, enabling reliable data capture, retrieval, and user-specific storage.
- Built a responsive React-based frontend to visualize physiological data, providing interactive daily and weekly charts generated from real-time sensor inputs.
- Integrated MAX30102 pulse oximeter sensor with a Particle Photon board, using synchronous state machines for stable and accurate signal processing.

- Designed the system architecture to support multiple concurrent users, ensuring real-time monitoring for several devices simultaneously without data loss or latency.
- Added configurable time intervals for data capture and implemented offline-first storage with automatic sync on reconnect, ensuring uninterrupted data availability.
- Conducted performance testing to validate system reliability and accuracy, supporting use cases in personal health monitoring and IoT-based healthcare solutions.
- **Skills:** Internet of Things (IoT), Node.js, Express, MongoDB, React, REST APIs, Data Visualization, Real-Time Charts

Cricket Match Result Forecasting | [GitHub](#) | [Website](#)

- Developed a Random Forest classifier to predict outcomes of ODI cricket matches, achieving 87% accuracy on historical match datasets.
- Built a real-time dashboard that updates match predictions every 5 minutes using simulated live match data, enabling dynamic analytics for fans and analysts.
- Designed and implemented backend APIs to serve dynamic match statistics and predictions based on historical performance, player stats, and team data.
- Enhanced user engagement by providing interactive visualizations and evolving match outcome predictions, encouraging exploration of analytics insights.
- Performed feature engineering and data preprocessing to improve model performance, including handling missing values, player form trends, and venue-specific factors.
- **Skills:** Random Forest Regressor, Python, Java, MySQL, API Design, Cricket Analytics, Real-Time Prediction, Data Preprocessing, Feature Engineering

Cricket Metrics | [GitHub](#) | [Website](#)

- Developed a dynamic cricket analytics dashboard using over 150,000 historical data points, providing comprehensive insights into player and team performance.
- Simulated live match data through backend APIs and integrated it with a Quarto-based frontend for real-time visualization.
- Created interactive plots including line, bar, donut, and map charts to visualize scoring patterns, player performance, and match trends.
- Built specialized APIs for batting, bowling, and partnerships data to support live match analysis and continuous updates.
- Implemented efficient data processing pipelines to handle large datasets, ensuring responsive and accurate dashboard performance.
- **Skills:** R, Shiny, Quarto, Java, MySQL, API Development, Data Visualization, ggplot2, Cricket Analytics, Real-Time Dashboards, Data Processing

Psychometric Analysis | [GitHub](#) | [Website](#)

- Conducted a psychometric analysis on 890 fictional characters across 100 universes using data from the Open-Source Psychometrics Project.
- Merged traits, MBTI types, and psychometric scores to study patterns in personality and character notability.
- Performed extensive data cleaning, feature engineering, and correlation analysis to uncover meaningful links between traits, audience engagement, and character prominence.
- Visualized insights with bar charts, density plots, and radar charts to highlight trends in personality traits and audience response.
- Developed interactive dashboards to allow exploration of relationships between MBTI types, psychometric scores, and character popularity.
- **Skills:** R, Quarto, Data Analysis, Data Visualization, ggplot2

COVID-19 Protocol Monitoring System | [GitHub](#) | [Published in IJRASET](#)

- Developed an active surveillance system integrating mask detection, social distance monitoring, temperature sensing, and automated sanitization to support COVID-19 safety protocols.
- Implemented mask detection using CNN with MobileNetV2 and OpenCV, achieving real-time face recognition and compliance checks.
- Designed a client-server architecture with Raspberry Pi Pico/Arduino Uno nodes connected via socket programming to a central server for scalable deployment across multiple entry/exit points.
- Integrated IR temperature sensors, motion detectors, and automated sanitizer dispensers for contactless entry management.
- Enhanced CCTV-based social distance monitoring by improving existing algorithms with YOLO object detection and COCO dataset, enabling real-time detection without recalibration for varying camera placements.

- Implemented SQLite database to log violations with timestamp, location, temperature, and captured image for centralized monitoring.
- Developed Flask-based web interface for front-end deployment, enabling access to reports and live surveillance feeds across devices in the same network.
- Published research findings in the International Journal for Research in Applied Science & Engineering Technology (IJRASET), highlighting the system's practical applicability and scalability.
- **Skills:** Computer Vision, CNN, MobileNetV2, YOLO, Socket Programming, Raspberry Pi, Arduino, Flask, OpenCV, SQLite, IoT

FBI Crime Data Reporting and NIBRS Adoption Rate Analysis | [GitHub](#)

- Analyzed 19,000+ U.S. law enforcement agency records from the FBI's Uniform Crime Reporting Program to study agency type distribution (City, County, Other) and participation in NIBRS reporting.
- Performed data preprocessing by imputing missing geographic coordinates, classifying agency types using text-based rules, and standardizing NIBRS participation start dates for temporal analysis.
- Conducted exploratory data analysis (EDA) with stacked bar charts, pie charts, line plots, and radial plots to visualize agency structures and NIBRS adoption trends across all 50 states.
- Identified significant differences in agency structures between urban and rural states, and highlighted uneven yet steadily increasing NIBRS participation with adoption spikes linked to policy shifts.
- Developed insights to support policymakers and law enforcement leaders in addressing lagging participation and advancing standardized crime data reporting practices.
- **Skills:** Python, Pandas, Data Cleaning, Exploratory Data Analysis (EDA), Data Visualization, Matplotlib, Seaborn

International Patients Treatment Management System | [GitHub](#)

- Developed a full-stack healthcare management system enabling international patient registration, treatment tracking, and cross-border case coordination.
- Implemented microservices architecture using Spring Boot and Spring MVC with Feign Client for efficient inter-service communication.
- Designed secure RESTful APIs to manage patient profiles, medical records, and billing workflows.
- Built a responsive Angular frontend dashboard with role-based interfaces for doctors, administrators, and patients.
- Integrated multi-currency billing and payment modules to streamline international financial transactions.
- Ensured data privacy and security through authentication, authorization, and role-based access control (RBAC).
- **Skills:** Java, Angular, Spring Boot, Spring MVC, Microservices, FeignClient, REST APIs, H2 Database

Flight Fare Prediction | [GitHub](#)

- Built a web application using Flask to predict flight fares based on input features like journey date, airline, source/destination, stops, and duration.
- Developed and trained a Random Forest regression model, achieving 95% prediction accuracy, and seamlessly integrated it into the application for real-time inference.
- Designed REST endpoints in Flask to accept user inputs, preprocess data, invoke the model, and return fare predictions dynamically.
- Created an intuitive UI with HTML/CSS templates for users to input travel details and view predicted prices.
- Handled feature engineering and preprocessing (datetime parsing, categorical encoding, duration splitting) to ensure accurate model input.
- **Skills:** Flask, Python, Scikit-learn, Pandas, Pickle, Web Development

EDUCATION

The University of Arizona

Masters in Science, Data Science — Distinguished Graduate Scholar

Tucson, AZ

Jan. 2024 – Dec. 2025

Savitribai Phule Pune University

Bachelor of Engineering, Computer

Pune, India

Aug. 2017 – May 2021

TECHNICAL SKILLS

Programming Languages: Python, R, Java, C++, C, JavaScript, SQL

Data Science & Machine Learning: NumPy, pandas, scikit-learn, XGBoost, Matplotlib, seaborn, Keras, TensorFlow, StatsModels

Development Practices: Object-Oriented Programming, Functional Programming, Microservice Architecture, MVC Pattern, Agile, Test-Driven Development, CI/CD

Databases & Querying: MySQL, SQLite, PostgreSQL, MongoDB, NoSQL

Cloud & DevOps: AWS, Docker

Natural Language Processing: spaCy, NLTK, Transformers (Hugging Face), fastcoref, Gensim, CoreNLP

Web & Backend Technologies: Angular, Spring Boot, Spring MVC, Node.js, Express.js, REST APIs, JWT, Maven, HTML, CSS

Tools & IDEs: Jupyter Notebook, PyCharm, VS Code, Android Studio, RStudio, IntelliJ, Git, GitHub, Postman, Docker Hub

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Plotly

Certifications: Amazon Web Services Certified Cloud Practitioner | [Credentials](#)

COURSEWORK

Core Data Science: INFO 511 – Foundations of Data Science, INFO 523 – Data Mining and Discovery, INFO 502 – Data Ethics

Machine Learning and NLP: INFO 557 – Neural Networks, INFO 539 – Statistical Natural Language Processing, INFO 555 – Applied Natural Language Processing

Data Engineering and Visualization: INFO 526 – Data Analysis and Visualization, INFO 531 – Data Warehousing in the Cloud

Web Development: ECE 513 – Web Development and the Internet of Things